

CS355 – Database Project Proposal Form

Section 1: Metadata

to be filled by the student

1.1. Project Information to be filled by the student

Title: CineFlix	
Section:	Instructor:

1.2. Student(s) Information

Name:	ID:
Section:	Batch:

Name:	ID:
Section:	Batch:

Name:	ID:
Section:	Batch:

Submission guideline: Save your project proposal as a pdf file and rename as Project Proposal_L1_ProposedTitle where L1 is to be replaced with your section

Section 2: The Project

to be filled by the student

2.1. Project Description: *Please provide a brief introduction of the project including its scope.*

This project focuses on developing an all-encompassing movie management system designed to serve multiple user roles, including regular viewers, premium viewers, administrators, and cinema managers. Each user type is granted tailored access and permissions based on their role, ensuring a customized experience. The primary objective is to create a dynamic and user-friendly platform where movie enthusiasts can seamlessly explore, search, filter, and review movie content. At the same time, administrators and cinema managers are empowered with efficient tools to update, manage, and curate the movie database. By addressing both user engagement and database management, the system aims to enhance the overall movie-viewing experience and streamline administrative operations in the entertainment industry.

2.2 Functional Requirements

This section describes each function/feature provided by your system. These functions are logically grouped into modules based on their purposes. The users in your system must be categorized such as client, customer or administrator etc. These users will be accessing the database with the level of access that they are authorized with.

Sample functional Requirements:

Module 1: Registrations

The system lets users register an account on the website as an administrator, manager and customer.

– Function 1a: **Register as a Viewer:** The register form prompts the user to enter their details i.e. Name, Email, Password. They will also be asked for location, which will be an optional field. The form is submitted and an unverified client account is created. The user receives a link on their email address which completes account verification.

– Function 1b: **Register as a Premium Viewer:** The register form prompts the user to enter their details i.e. Name, Email, Password. They will also be asked for location, which will be an optional field. The user is then redirected to another page on which they add their payment details. The form is submitted and an unverified client account is created. The user receives a link on their email address which completes account verification.

– Function 1c: **Register as an administrator:** The register form prompts the user to enter their details i.e. Name, Email, Password, and other information. The form is submitted and an unverified customer account is created. The user receives a link on their email address which completes account verification.

– Function 1d: **Register as a cinema manager:** The register form prompts the user to enter their details i.e. Name, Email, Password, and other information. The form is submitted and an unverified customer account is created. The user receives a link on their email address which completes account verification.

Module 2: Login

Users would login as either viewers, administrator or cinema managers. Depending on the role, they would interact with varying interface; viewers would be able to search/view movies, filter movies according to their choice and leave a review. Administrators can add/delete movies, reviews/ratings and edit descriptions. Cinema managers can add/delete/edit movies streaming in the cinema. Both administrators and cinema managers can view and search movies.

- Function 1a: **Viewers** – Login form would ask for user ID and password (it would accordingly detect whether it belongs to a viewer and hence show the required interface). For security purposes, a confirmation link would be sent to the associated email ID.

- Function 1b: **Administrator** - Login form would ask for user ID and password (it would accordingly detect whether it belongs to an admin and hence show the required interface). For security purposes, a confirmation link would be sent to the associated email ID.

- Function 1c: **Cinema managers** - Login form would ask for user ID and password (it would accordingly detect whether it belongs to a manager and hence show the required interface). For security purposes, a confirmation link would be sent to the associated email ID

- Function 2: **ALL** – There will be a forgot password link which would send an email to the associated email ID to start a resetting process

MODULE 3: SEARCH/ADD/DELETE

- Function 1: All Viewers – They can input a movie title into the search bar to find specific films.

- Function 2: All Viewers – They can click on a movie title displays its details, including title, producer, ratings, IMDB rating, duration, and genre. A link to the movie's reviews is also provided.

- Function 3: All Viewers – They can add movies to their Wishlist for future reference or potential viewing.

- Function 4: Premium Viewers – Premium users have the privilege of downloading movies for offline viewing.

- Function 5: Administrator – They can add new movies to the database, including all relevant details. They can also add initial reviews to the database.

- Function 6: Administrators – They have access to modify movie details and user reviews as needed. This includes updating information and correcting errors.

- Function 7: Administrators - They can also adjust user subscription statuses, switching between premium and regular levels as requested by the users.

- Function 8: Administrator – They can delete a movie from the movie database with all its details.

- Function 9: Cinema managers - They can add movies to their cinema's schedule, specifying details such as showtimes, pricing, and any special screenings.

-Function 10: Cinema managers – They can remove movies from their cinema's schedule, typically when the movie's run has ended or due to other reasons.

-Function 11: Administrator – If a movie is playing in a cinema located in the viewer's city, the administrator will update the movie database by marking the "Being Screened" column as "Yes." Otherwise, it will be marked "No".

MODULE 4: FILTER

This advanced feature allows users to refine their movie search using various criteria, including genre, production country, duration, ratings, release year, director, and whether the movie is currently showing in a nearby cinema. Users can apply one or more filters to narrow down their results. The system will then display a list of movies that match the specified criteria. If the user searches for a movie that's playing in a cinema, the system will also provide information about the nearby theaters where it's showing.

MODULE 5: REVIEW & FEEDBACK

- Function 1: Viewers – for movies they have watched, viewers would be able to rate the movies. This rating would be on a scale of 1-10 with 1 being worst and 10 being best.

-Function 2: Administrator – admin would be able to view the reviews and can choose to delete a movie constantly getting bad reviews.

-Function 3: Viewers – If they like a movie, they can mark it as a favorite so their movie suggestions will show similar movies.

2.3. Planned Schedule: *Kindly list the start/end dates and the timeline for the achievement of any intermediate milestones and the expected contribution to be made by the participant(s).*

We will work together according to the following schedule:

- By week 5 – Make ERD
- By week 7 – Formulate dummy data
- By week 9 – Implement SQL
- By week 11 – Complete GUI and ERD
- By week 11-13 – Functionality
- By week 14 – Finalize project
- By week 15 – Present

2.4. Technology Stack: *If you are utilizing any language or database besides PyQt and SQL Server, please complete this section; otherwise, leave it blank. Specify the programming language and database management system intended for constructing this application, as well as the application type (Desktop, Web, or Mobile).*

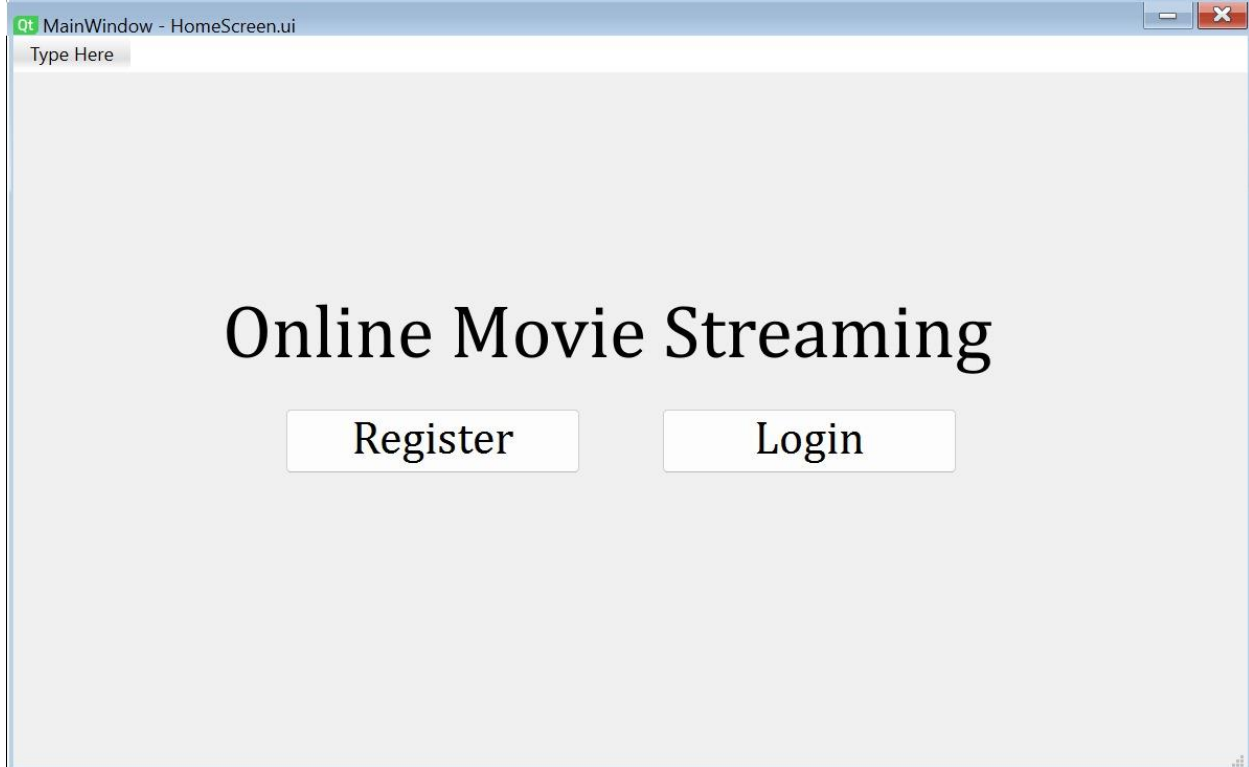
SOFTWARES:

1. QTdesigner
2. MySQL
3. Visual Studio
4. Visual Studio Code
5. GitHub

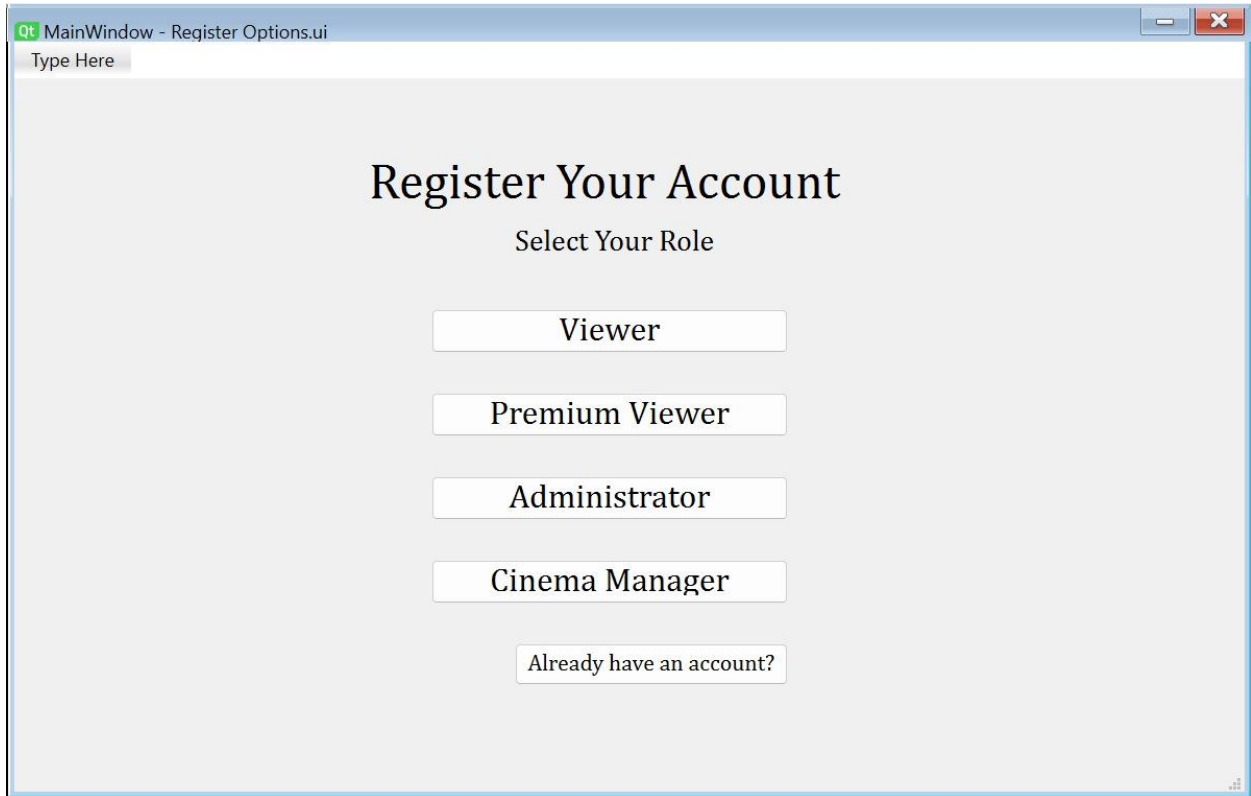
LANGUAGES:

1. C#
2. Python
3. SQL

2.5. Screens: Provide images of all application screens, showcasing clear input and corresponding outputs. Ensure each image includes a concise caption explaining user action and expected/observed output. You can create these screens using Qt Designer.



HomeScreen that the user interacts with for the first time. The user can either register their account or Login (if they already have an account made)



Qt MainWindow - Register Options.ui

Type Here

Register Your Account

Select Your Role

Viewer

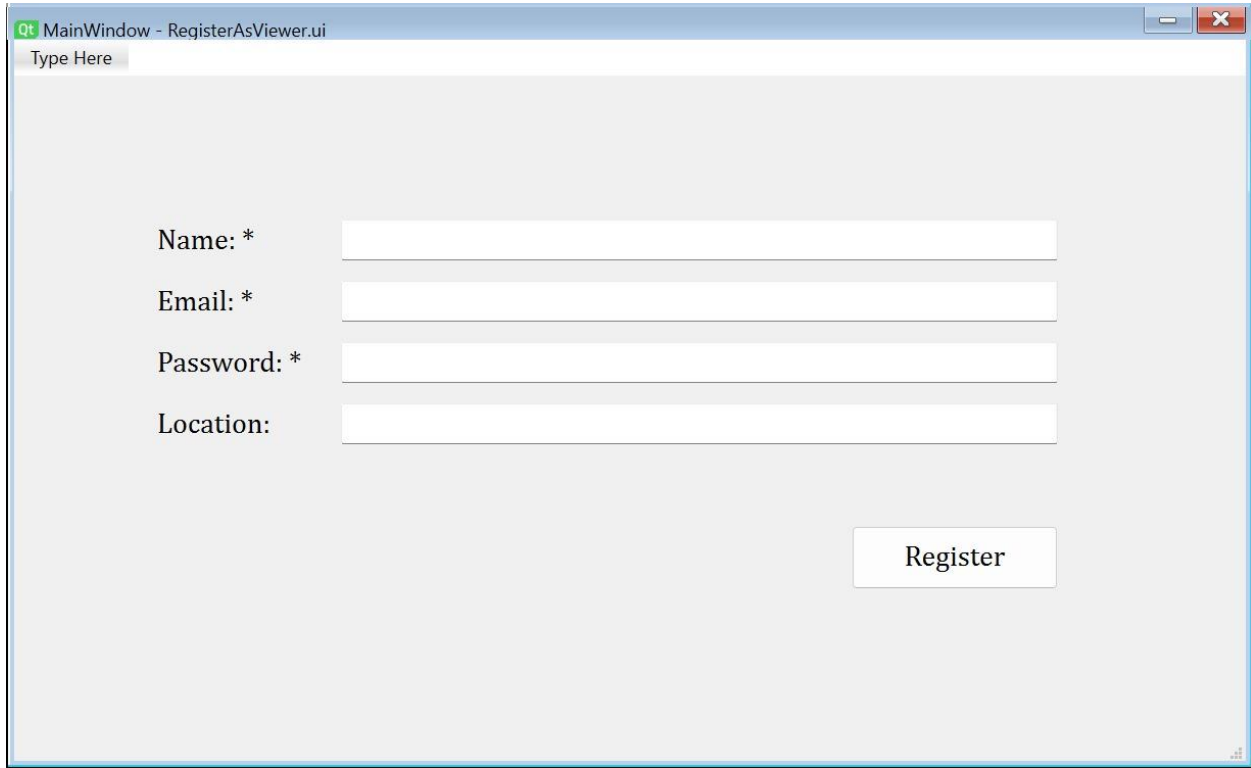
Premium Viewer

Administrator

Cinema Manager

Already have an account?

If the user clicks on Register, they get to select their role



Qt MainWindow - RegisterAsViewer.ui

Type Here

Name: *

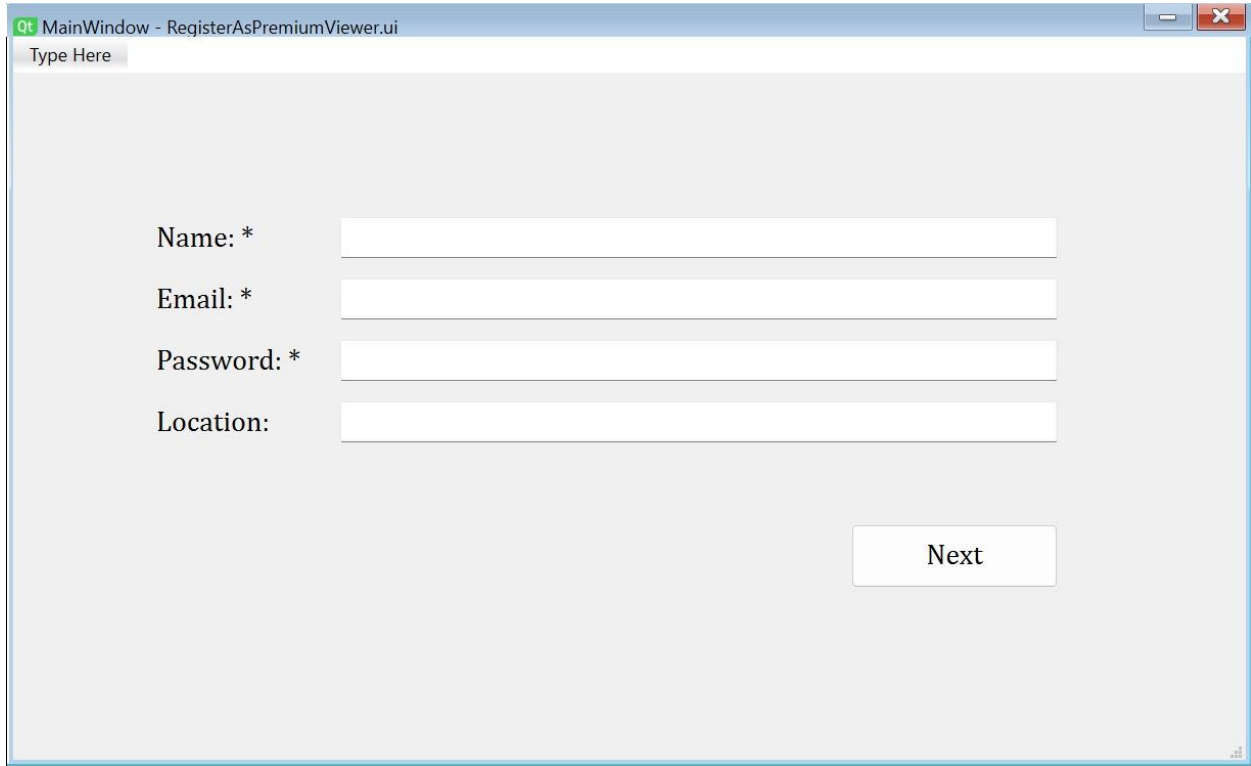
Email: *

Password: *

Location:

Register

If the user selects the role of a viewer, they get directed to this window and enter information to register



Qt MainWindow - RegisterAsPremiumViewer.ui

Type Here

Name: *

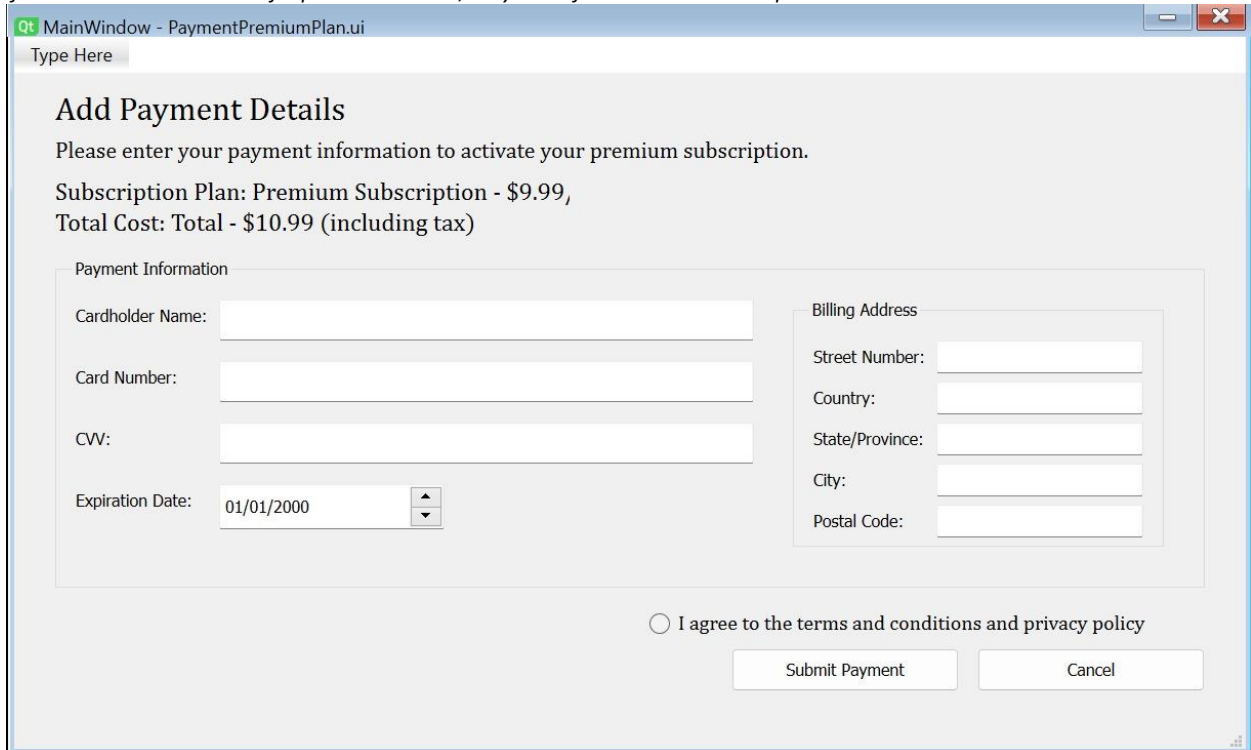
Email: *

Password: *

Location:

Next

If the user selects the role of a premium viewer, they must fill in the details and press next



Qt MainWindow - PaymentPremiumPlan.ui

Type Here

Add Payment Details

Please enter your payment information to activate your premium subscription.

Subscription Plan: Premium Subscription - \$9.99,
Total Cost: Total - \$10.99 (including tax)

Payment Information

Cardholder Name:

Card Number:

CW:

Expiration Date:

Billing Address

Street Number:

Country:

State/Province:

City:

Postal Code:

☐ I agree to the terms and conditions and privacy policy

Submit Payment Cancel

After clicking next, they get directed to the window for payment details

Qt MainWindow - RegisterAsAdministrator.ui

Type Here

Name: *

Email: *

Password: *

Job Title:*

Department:*

Office Location:*

Experience Level:*

If the user selects the role of an Administrator, they get directed to this window and enter information to register

Qt MainWindow - RegisterAsCinemaManager.ui

Type Here

User Information

Name: *

Email: *

Password: *

Phone Number:*

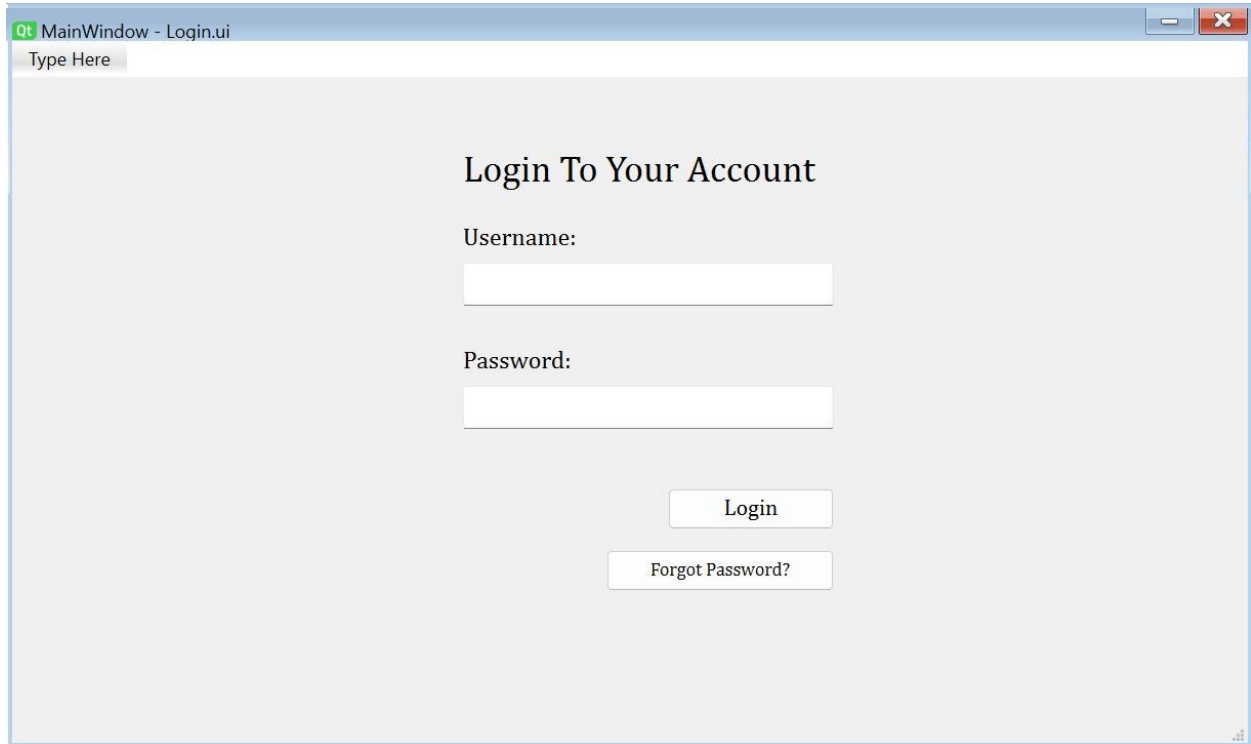
Experience:*

Cinema Information

Cinema Name: * Capacity: *

Location:*

If the user selects the role of an Cinema Manager, they get directed to this window and enter information to register



Qt MainWindow - Login.ui

Type Here

Login To Your Account

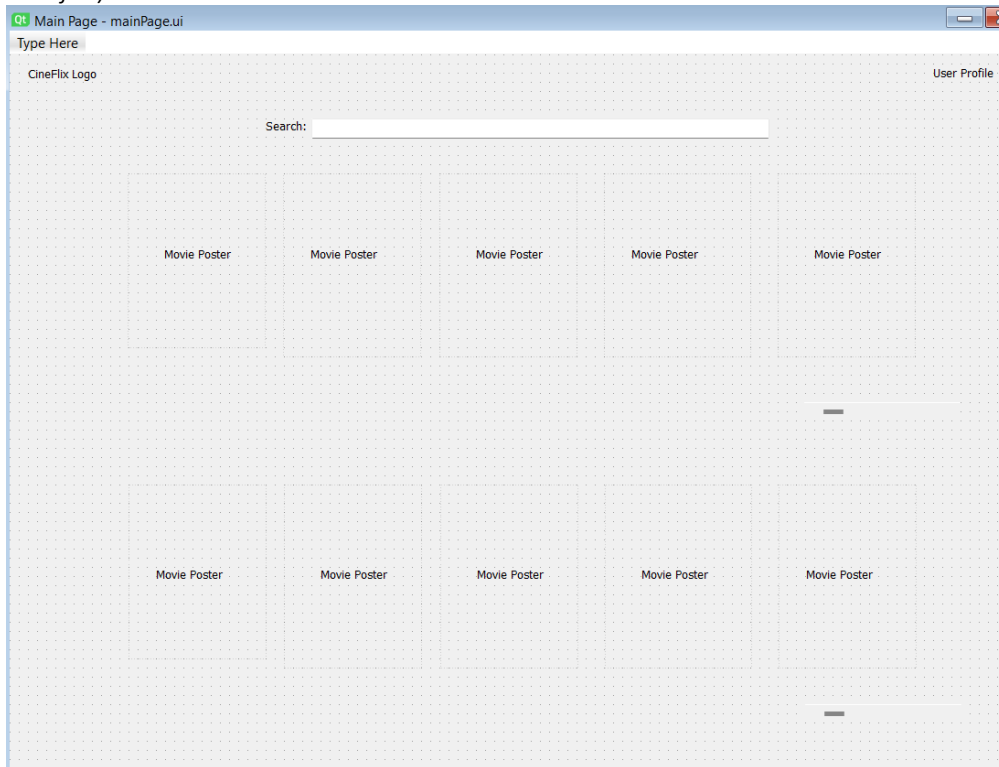
Username:

Password:

Login

Forgot Password?

If the users have already registered, they can login after entering the username and password (the roles will be automatically identified)



Qt Main Page - mainPage.ui

Type Here

CineFlix Logo

User Profile

Search:

Movie Poster

Movie Poster

Movie Poster

Movie Poster

Movie Poster

Movie Poster

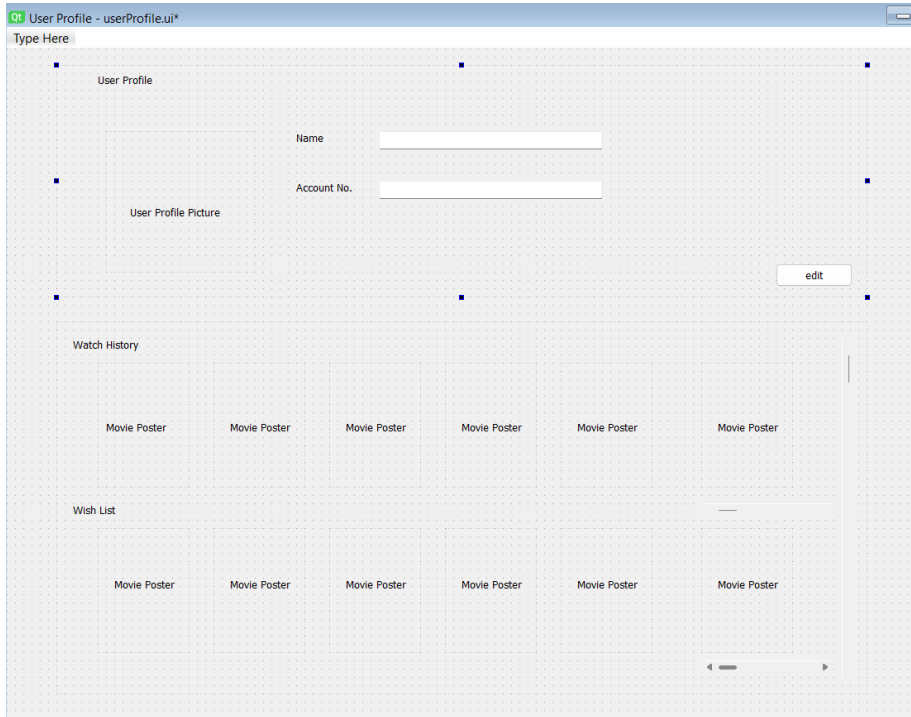
Movie Poster

Movie Poster

Movie Poster

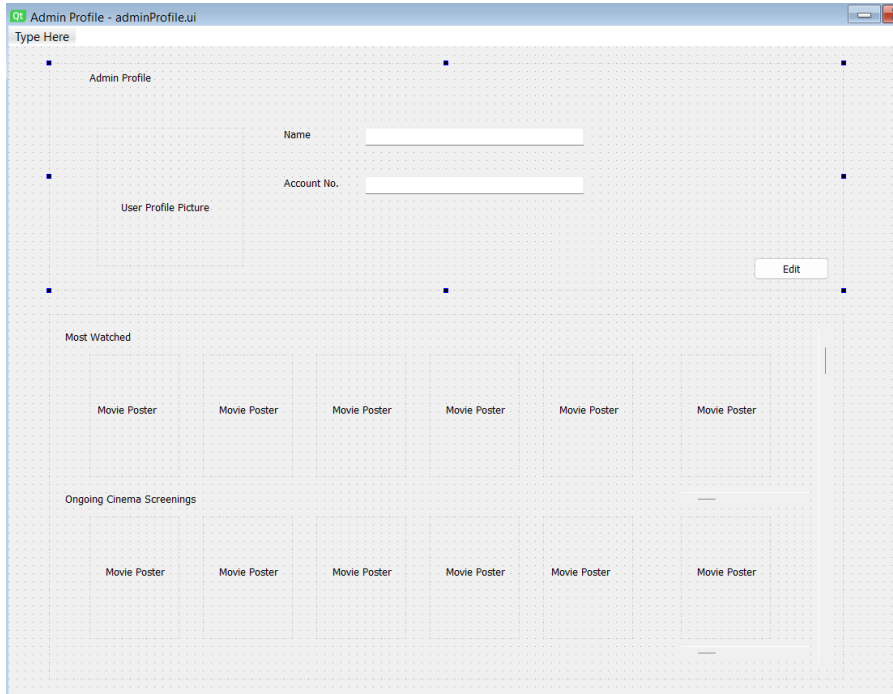
Movie Poster

Main Screen after Login



The screenshot shows a Qt Designer window titled "User Profile - userProfile.ui*". The main widget is a grid layout. At the top left is a label "User Profile". Below it is a "User Profile Picture" placeholder. To the right of the picture are two text input fields: "Name" and "Account No.". An "edit" button is located to the right of the "Account No." field. Below the profile section is a "Watch History" section, which contains a horizontal scroll area with six "Movie Poster" placeholders. Below that is a "Wish List" section, also containing a horizontal scroll area with six "Movie Poster" placeholders.

User Profile Page



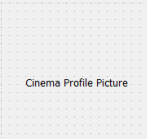
The screenshot shows a Qt Designer window titled "Admin Profile - adminProfile.ui". The main widget is a grid layout. At the top left is a label "Admin Profile". Below it is a "User Profile Picture" placeholder. To the right of the picture are two text input fields: "Name" and "Account No.". An "Edit" button is located to the right of the "Account No." field. Below the profile section is a "Most Watched" section, which contains a horizontal scroll area with six "Movie Poster" placeholders. Below that is an "Ongoing Cinema Screenings" section, also containing a horizontal scroll area with six "Movie Poster" placeholders.

Admin Profile Page

Qt Cinema Manager Profile - cinemaManagerProfile.ui*

Type Here

Manager Profile



Cinema Name

Account No.

[Edit](#)

Current Screenings

Movie Poster	Movie Poster	Movie Poster	Movie Poster	Movie Poster	Movie Poster
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Demand:

Movie Poster	Movie Poster	Movie Poster	Movie Poster	Movie Poster	Movie Poster
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Cinema Manager Profile Page